

Panel (c) considers incumbent workers—those employed at the workplace before the arrival of AI chatbots—whose labor market outcomes we can track regardless of whether they remain with or separate from their original firm.²² These workers have not experienced differential changes in hours. Figure F.9 shows that they have also not experienced differential occupational mobility. Hence, while Section 3.3 shows that workers’ use of AI chatbots has helped them move *into* new occupations, employers’ adoption of AI chatbots has not displaced workers *out of* their original occupations.

Panel (d) examines overall rates of job creation and destruction, showing that job churn at workplaces that encourage AI chatbots has remained stable, with confidence bands ruling out effects on monthly job churn rates greater than 0.5% of employment.

Heterogeneity by Intensity of Adoption. Figure 6, Panels (c) and (d) examine the effects of additional initiatives in enterprise chatbots and training. Although Section 2.2 shows that uptake and reported benefits are highest when the full suite of initiatives is implemented, we find no detectable impacts on overall employment or early-career jobs at these workplaces. Similarly, Appendix F.4 shows that incumbent workers and job churn have not changed differentially at workplaces with these investments.

Occupational Heterogeneity. Figure 7, Panels (c) and (d) report the estimated effects of AI chatbots on workplace employment and early-career jobs across the 11 occupations in our sample. Confidence bands are again wide for these occupation-specific estimates and should therefore be interpreted with caution. Nonetheless, we find no statistically significant effects in any occupation, and point estimates are generally near zero.²³ Even between our most contrasting groups—software developers and marketers (with flexible labor rules) versus teachers (with collective bargaining)—we observe no systematic differences in employment impacts. Moreover, while Appendix D.2 documents aggregate declines in early-career jobs in several of our exposed occupations, including

²²This analysis of incumbent workers resembles the research designs of Autor et al. (2014); Walker (2013).

²³Appendix F.4 similarly shows null results for job churn and worker sorting.

software development, customer support, legal professions, and marketing, these declines are not driven by firms adopting generative AI chatbots. This suggests that other forces, such as changes in the macroeconomic environment or anticipatory hiring freezes unrelated to actual adoption, may be driving the aggregate pattern.

Discussion. Workplaces that have adopted AI chatbots have not fared differently in terms of employment, early-career jobs, job churn, or incumbent worker outcomes. How do we reconcile this “still surface” with the “rapid currents” documented in Section 2? The task creation evidence in Section 2.3 provides a key part of the answer: employers are absorbing generative AI through task reorganization—creating new roles, reassigning responsibilities, investing in training and oversight—rather than adjusting on the margin of total labor input. Consistent with this, workers who report larger time savings are also more likely to have taken on new tasks (Appendix Figure E.2), suggesting that individual productivity gains are absorbed into task reorganization. In this sense, the null employment results are not evidence of inaction but of a specific mode of adjustment: one that operates through the internal reorganization of work rather than through the external labor market. This pattern is consistent with the productivity J-curves observed at the onset of prior general-purpose technologies, where substantial organizational investments precede measurable changes in economic outcomes (Brynjolfsson, Rock and Syverson, 2021; David, 1990; Solow, 1987).

3.5 Perceived Impacts

To complement the difference-in-differences analysis, we asked respondents directly: “*Have AI chatbots affected your labor earnings?*” If so, we followed up with, “*By how much?*” If AI chatbots were meaningfully affecting labor market conditions, workers in exposed occupations would likely notice.

Appendix E.5 reports workers’ perceived earnings impacts from AI chatbots. Virtually all respondents—97.7% of adopters and 99.5% of non-adopters—state that AI chatbots have had *no* impact on their labor earnings. While individuals may struggle to assess

such counterfactuals precisely, these responses are informative in three ways.

First, the fact that virtually all non-adopters report no earnings impact suggests that occupation-wide spillovers are minimal. Despite broad adoption and changes in work content, AI chatbots do not appear to have shifted equilibrium wages or hours—reinforcing the conclusion from our identification analysis in Section 3.1 that the “missing intercept” remains small.

Second, even adopters report small average earnings impacts, ranging from 0% to 0.5% depending on occupation and employer chatbot initiatives. These estimates are an order of magnitude smaller than the time savings reported in Section 2.2 and fall within the confidence intervals of the difference-in-differences estimates for earnings in Section 3.2.

Third, the limited average earnings impacts primarily reflect the overwhelming share of workers reporting no change—rather than large positive and negative effects offsetting each other. This supports the conclusion that AI chatbots have not had significant heterogeneous impacts on workers’ earnings.

Taken together, these self-reported perceptions from workers in the most exposed occupations reinforce our finding that AI chatbots have so far had limited impacts on earnings in the labor market.

4 Conclusion

This paper studies how labor markets adjust at the onset of major technological change, providing large-scale evidence on the early impacts of AI chatbots—the most widely adopted generative AI tool to date. By linking extensive surveys on AI chatbot adoption and work organization in 11 exposed occupations to administrative labor market records in Denmark, we document both what moves and what does not.

What moves is the structure of work. Most employers in exposed occupations have implemented chatbot initiatives, workers report substantial productivity benefits, and new AI-related tasks have emerged broadly—especially in workplaces that encourage and invest in the tools. These patterns are consistent with task-based theories of technology adoption

(Acemoglu and Restrepo, 2018a) and with evidence that complementary organizational investments are critical to realizing the productivity gains from new technologies (Bloom, Sadun and Reenen, 2012; Brynjolfsson and Hitt, 2000). Occupational mobility is the one margin where these rapid currents break through to administrative labor market outcomes: workers who use AI chatbots are more likely to have transitioned into new occupations, particularly into roles where workers have greater discretion over their toolsets. These workers move into better-paid occupations where using AI chatbots is more relevant. This supports the view that generative AI may help workers access otherwise scarce expertise, with broader implications for inequality in the labor market (Autor, 2024; Autor and Thompson, 2025).

What does not move is the surface—recorded hours and earnings. Using difference-in-differences, we estimate precise null effects at both the worker and workplace levels, ruling out effects larger than 2% two years after ChatGPT’s launch. These null results hold across occupations, across intensities of adoption, and even among workers who report substantial daily time savings. While aggregate early-career employment has declined in several exposed occupations, firms’ adoption of AI chatbots does not drive these declines.

We conjecture that these two patterns may be mutually reinforcing: when earnings and recorded hours remain stable, workers and workplaces absorb technological change through task reallocation and other unrecorded adjustments, reducing pressure on pay or hours. If so, the absence of measurable labor market effects is not evidence that nothing is happening—but rather that transformation is occurring on margins that conventional economic statistics do not capture. This pattern echoes the productivity J-curves observed at the onset of prior general-purpose technologies (Brynjolfsson, Rock and Syverson, 2021; David, 1990; Solow, 1987), and our data reveal what lies within the trough: the reorganization of work that absorbs technological change before it surfaces in hours or earnings. Understanding this process requires the kind of linked data infrastructure we develop here, capturing both the reorganization of work within firms and its translation into administrative labor market outcomes.